

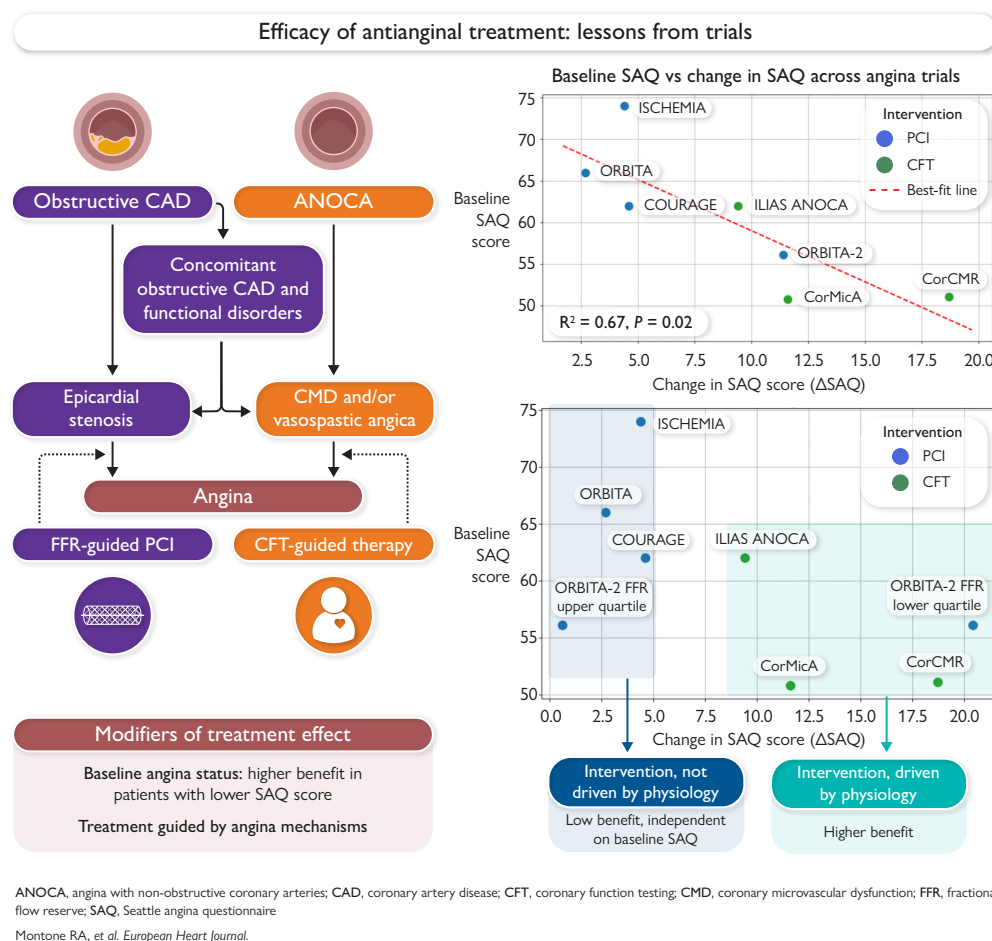
Treatment of angina pectoris: lessons from trials

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Graphical Abstract



Left panel. Mechanisms of angina and modifiers of treatment effect. **Upper right panel.** Baseline SAQ vs SAQ change across trials. We reported SAQ at 6-months as time point for all trials, except for ORBITA (6-weeks) and ORBITA-2 (12-weeks). We reported SAQ-SS when available (ISCHEMIA, CorMicA, ILIAS-ANOCA, CorCMR), otherwise we calculated SAQ-SS by averaging the individual domain scores for angina frequency, physical limitation, and quality of life (COURAGE and ORBITA-2) or angina frequency, physical limitation, and angina stability (ORBITA). **Lower right panel.** Baseline SAQ vs SAQ change across trials according to the implementation of physiology. ORBITA-2 Lower FFR quartile: FFR < 0.46; upper FFR quartile: FFR > 0.73. CFT, coronary function testing; CMD, coronary microvascular dysfunction; FFR, fractional-flow reserve; SAQ, Seattle Angina Questionnaire.

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Introduction

The management of stable angina pectoris has undergone a profound re-evaluation over the past two decades. Longstanding practice patterns, rooted in the logic that relieving an epicardial stenosis should improve myocardial ischaemia and thus symptoms, have been tested in several modern randomized trials. For patients with stable obstructive coronary artery disease (CAD), PCI has been shown to reduce the amount of myocardial ischaemia, but its impact on symptoms is more nuanced. The incremental benefit of PCI, beyond what is achieved with optimal medical therapy (OMT) alone, is often modest when viewed at the population level, though certain symptomatic subgroups may derive greater relief.¹⁻⁵

Meanwhile, an equally important but historically neglected group—patients with angina in the absence of obstructive coronary stenoses (ANOCA)—has come into focus. Far from being ‘normal’, many of these patients suffer from coronary vasomotor disorders including microvascular angina, vasospastic angina, or mixed forms of coronary dysfunction.^{1,6} The implementation of a coronary function testing (CFT) allows defining the precise ANOCA endotype, and when therapy is tailored to the underlying mechanism, meaningful improvement in angina status is achievable, even more pronounced than benefit derived from PCI in patients with obstructive CAD.⁶⁻⁹ Yet, in everyday practice, CFT remains underutilized, and empirical management still predominates.

In this context, understanding the expected magnitude of symptom improvement from different interventions is essential both in patients with obstructive CAD and in ANOCA.

Measuring angina outcomes: the Seattle angina questionnaire

The Seattle Angina Questionnaire (SAQ) is a disease-specific, self-administered instrument designed to capture the patient's perspective on their angina.¹⁰ It evaluates multiple domains—physical limitation, angina frequency, angina stability, quality of life, and treatment satisfaction—each scored from zero to one hundred, with higher scores representing better health status. Among these, the most frequently used composite measure in trials is the SAQ Summary Score (SAQ-SS), which averages the physical limitation, angina frequency, and quality of life domains.

Interpretation of SAQ changes requires an understanding of what constitutes a clinically important difference. A shift of ~5 points is generally considered small, a change of 10–20 points is moderate and a gain exceeding 20 points is large and often transformative for the patient's daily experience.¹⁰

Angina and PCI in obstructive CAD: who really benefits?

The modern evidence base for angina relief by PCI in stable obstructive CAD is anchored in trials such as ISCHEMIA, COURAGE, FAME-2, ORBITA and ORBITA-2.

The COURAGE trial was the first large study to challenge the revascularization dogma,³ randomizing patients to PCI plus OMT or OMT alone, with a median follow-up of 4.6 years. COURAGE demonstrated no difference in death or myocardial infarction, but in terms of symptoms, PCI offered early relief:

at 1 year, about 66% of patients in the PCI group were free of angina compared with 58% in OMT group. However, at 3-year the difference narrowed, and at 5-year it disappeared. The SAQ was administered in a subset of patients, showing modest short-term benefits of PCI, typically in the range of 4–6 points in SAQ-SS, diminishing with time. Importantly, patients with frequent baseline angina derived greater benefit, while minimally symptomatic patients gained little.

The FAME-2 trial randomized patients with stable CAD to fractional-flow reserve (FFR)-guided PCI plus OMT or OMT alone, demonstrating a significant reduction in urgent myocardial revascularization and a lower occurrence of angina Canadian Cardiovascular Society (CCS) II to IV grade at 1-year in the FFR-guided PCI group,¹¹ thus suggesting a benefit deriving from a physiology-guided revascularization.

The ISCHEMIA trial provided the most contemporary large-scale evidence.³ Patients with moderate-to-severe ischaemia were randomized to an initial invasive strategy (PCI or CABG, plus OMT) vs a conservative strategy. At baseline, about half the cohort reported monthly angina, and 20% reported weekly or daily angina. At 3-month, the invasive arm showed a mean SAQ-SS improvement of +4.2 points over the conservative group, with the benefit concentrated among those with daily or weekly symptoms, where improvements reached 8–10 points. At 12-months, the between-group difference remained significant, but it narrowed, and at 36-months became non-significant.

The ORBITA trial provided an even more stringent test by enrolling patients with single-vessel CAD in a randomized, double-blind, sham-controlled design.⁴ After 6-weeks of intensive antianginal therapy, patients were randomized to PCI or sham procedure. At baseline, most participants reported at least weekly angina. After 6-weeks, PCI improved exercise time by 16.6 s over sham, a modest effect, and the placebo-adjusted treatment effect on the SAQ angina frequency domain was only +2.5 points, with no significant change in SAQ-SS. These findings underscore both the limited symptomatic benefit of PCI in relatively well-treated patients and the substantial contribution of placebo effects. Finally, the ORBITA-2 trial demonstrated that in patients with stable angina who were not receiving antianginal medication, PCI resulted in a significantly lower angina symptom score than a placebo procedure, with +11.4 points in SAQ-SS at 12-weeks.¹²

Taken together, these trials demonstrate a consistent pattern: PCI may improve angina, but in patients receiving OMT it produces only modest average gains in SAQ scores, typically 3–5 points, with benefits amplified in patients with high baseline symptom burden but negligible in those with few symptoms. The effect is largely confined to the first year and wanes over time.

Coronary function testing in ANOCA: unmasking the mechanism

ANOCA represents a heterogeneous group in which angina arises from microvascular dysfunction, vasospasm, or both.¹ These pathophysiological mechanisms are not detectable at coronary angiography and would be missed without the implementation of CFT.^{1,2}

The CorMicA trial provided the first randomized evidence of this approach.⁷ A total of 151 ANOCA patients underwent

comprehensive invasive CFT, including acetylcholine testing and measurement of CFR and IMR. In the intervention arm, results were disclosed, and therapy tailored accordingly, while in the control arm results were withheld and patients received standard care. At baseline, mean SAQ-SS was 49, reflecting marked symptomatic impairment. At 6-month, the intervention group improved their SAQ-SS by +11.6 points compared with controls.

The ILIAS ANOCA trial confirmed these findings,⁸ randomizing patients to routine ad-hoc invasive CFT with targeted therapy or to standard care without CFT. At 6-month, the intervention arm demonstrated a +9.4-points in SAQ-SS compared with controls. Notably, the diagnostic yield for a vasomotor disorder was 78%, underscoring the high prevalence of treatable coronary dysfunction in this population.

Finally, the recent CorCMR trial demonstrated that non-invasive endotyping with adenosine stress perfusion cardiac magnetic resonance imaging (MRI) followed by tailored therapy portended a +20.9 points in SAQ-SS at 12-month compared with controls not receiving MRI after coronary angiography.⁹

These studies consistently show that CFT-guided therapy delivers improvements in angina status in ANOCA, with effect sizes that exceed those typically observed for PCI in unselected stable obstructive CAD.

From mechanism to therapy: why knowledge of pathophysiology shapes angina outcomes

The key to reconciling these findings lies in two critical aspects: (i) patient selection; and (ii) comprehension of the mechanism of angina (*Graphical Abstract*). Across trials, the benefit of PCI closely follows baseline symptom burden. In ISCHEMIA, a trial designed to enrol patients with ischaemia (not necessarily angina), 35% of patients had no history of angina during the previous 4-week, and 10% had no history of angina at all. Thus, minimally symptomatic patients gained little from PCI, whereas those with frequent angina experienced greater relief. On average, PCI provides only modest improvements in angina status in modern cohorts, largely because many patients have infrequent symptoms and receive OMT. For patients with frequent angina, PCI can be impactful, but they are a minority in stable CAD populations.¹⁻⁵

However, even more important is the comprehension of angina mechanisms. Patients with ANOCA often suffer persistent angina despite no obstructive lesions. Without CFT, they are treated empirically, often with mismatched therapies, leading to partial or absent relief.^{6, 7} ANOCA trials have shown that identifying the underlying mechanism enables targeted therapy and meaningful improvement in symptoms.⁷⁻⁹ The same applies to obstructive CAD, whereby PCI will not improve symptoms if the stenosis is not the main driver. Many patients with obstructive CAD also have microvascular dysfunction or vasospasm, either of which can dominate the clinical picture.^{13,14} Studies suggest that 30%–50% of such patients continue to have angina after successful PCI due to coexisting microvascular disease.^{1,13,14} The FAME-2 and ORBITA-2 trials further refined this understanding, showing that physiology-guided PCI predicts symptomatic benefit, with patients having lower FFR

values gaining the most,^{11,15} and the implementation of CFT with microvascular assessment can help to identify the true ischaemic substrate and prevent unnecessary PCI in patients whose symptoms originate from non-epicardial mechanisms.¹⁴

Implications for clinical practise are considerable. First, symptom relief following PCI in CCS should be regarded as probabilistic rather than guaranteed when the underlying angina mechanism remains undefined. Second, ANOCA is neither benign nor trivial—it is prevalent, burdensome, and frequently amenable to targeted therapy when systematically evaluated. Third, diagnostic pathways should therefore incorporate functional assessment—whether through invasive CFT or validated non-invasive surrogates—at an earlier stage for both ANOCA and obstructive CAD.^{2,16} Practical implementation could involve embedding standardized CFT protocols within PCI-capable catheterization laboratories, enabling concurrent anatomical and physiological assessment. Establishing multidisciplinary heart teams, including interventional cardiologists, imaging specialists, and dedicated angina services, may further support comprehensive evaluation and longitudinal management. Finally, the emergence of non-invasive CFT surrogates, including cardiac MRI, computed tomography (CT)-derived perfusion coronary angiography, and positron emission tomography, may offer promising avenues to extend accessibility across clinical settings,^{2,9,16} even if non-invasive techniques may not capture the full spectrum of vasomotor disorders, particularly vasoconstriction disorders or mixed forms.

From a policy and systems perspective, broader integration of CFT requires overcoming practical barriers related to operator training, laboratory workflow, reimbursement, and clinical awareness. Yet the high prevalence of ANOCA and the magnitude of patient benefit make investment worthwhile. At the same time, the continued overuse of PCI for symptom relief in stable CAD, without clarifying ischaemic mechanisms, risks disappointing patients and diverting resources from more effective diagnostic strategies.

Conclusions

Stable angina is a heterogeneous condition, and symptom relief depends on matching treatment to the mechanism of ischaemia. In unselected obstructive CAD, PCI offers only modest average benefit, with greater gains when guided by functional assessment. For ANOCA, CFT-guided therapy often provides larger improvements than PCI in stable CAD, and underuse of CFT in everyday clinical practise represents a major missed opportunity. Therefore, clinicians should prioritize mechanism-based diagnosis and therapy, ensuring that angina management is both physiologically sound and patient-centred, as underscored in current ESC CCS Guidelines.²

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Declarations

Disclosure of Interest

All authors declare no disclosure of interest for this contribution.

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